

Name: vrajesh

Wide Area Networks

1. Draw a simple diagram (on paper or using an online tool like draw.io) showing how your phone connects to Instagram's servers from your home WiFi, including the ISP and at least two Wide Area Network (WAN) devices (like routers or switches) in the path.

Answer:

Diagram

...

Instagram Server



Internet (WAN)



ISP Core Router



ISP Distribution Switch



Home WiFi Router



Mobile Phone 

...

Explanation:

*When you open Instagram on your mobile phone, your phone first connects to your ****Home WiFi Router****.*

*The router sends your request to your ****Internet Service Provider (ISP)****. The ISP forwards the request through its ****Core Router**** and ****Distribution Switch****, which are WAN devices used to move data over long distances. The request then travels through the Internet (WAN) and reaches the ****Instagram Server****. The server processes your request and sends the response back through the same path until it reaches your phone.*

2. Create a table listing three types of WAN technologies (such as MPLS, Leased Line, and Broadband Internet) and compare them based on speed, cost, and where you might see them used in real life (e.g., Zomato office, IRCTC ticketing, your home).

Answer:

MPLS (Multiprotocol Label Switching)

** **Speed:** High*

** **Cost:** Expensive*

** **Used In:** Large companies like Zomato offices, banks, and corporate organizations.*

** **Explanation:** MPLS provides a fast, secure, and reliable connection between different office locations. It is mainly used by businesses that need stable communication.*

Leased Line

** **Speed:** Very High (Dedicated Connection)*

** **Cost:** Very Expensive*

** **Used In:** IRCTC ticket booking systems, hospitals, data centers, and large companies.*

** **Explanation:** A leased line provides a dedicated internet connection that is not shared with other users. It offers excellent speed and reliability for important business operations.*

Broadband Internet

** **Speed:** Medium to High*

** **Cost:** Affordable*

** **Used In:** Homes, schools, colleges, and small businesses.*

** **Explanation:** Broadband is the most common internet connection used by home users. It provides good speed at a low cost and is suitable for everyday activities like browsing, online classes, and video streaming.*

3. Research and write a short explanation of how a VPN (Virtual Private Network) uses WANs to let you access apps like Spotify or YouTube securely from a public WiFi at a coffee shop.

***Hint:** Focus on how the VPN creates a private tunnel over the public WAN.*

Answer:

*A VPN (Virtual Private Network) allows you to use the internet securely, even when connected to public WiFi. When you connect to a VPN, it creates a ****private and encrypted tunnel**** between your device and the VPN server over the public Wide Area Network (WAN).*

For example, if you are using free WiFi at a coffee shop to access Spotify or YouTube, your internet traffic first goes through the encrypted VPN tunnel instead of directly to the website. Because the data is encrypted, hackers or other people using the same WiFi cannot read or steal your personal information. The VPN server then forwards your request to Spotify or YouTube and safely returns the response back to your device through the same secure tunnel.

Using a VPN helps protect your passwords, personal information, and browsing activity while using public internet connections.

4. Imagine Flipkart wants to connect its warehouses in Ahmedabad, Mumbai, and Delhi. Suggest which

WAN technology would be best for them and justify your answer with two reasons.

Answer:

*The best WAN technology for connecting Flipkart warehouses in ****Ahmedabad, Mumbai, and Delhi**** is ****MPLS (Multiprotocol Label Switching).*****

Reason 1:

*MPLS provides a ****fast and reliable network connection**** between different warehouse locations. This allows employees to quickly share inventory information, customer orders, and delivery updates without delays.*

Reason 2:

*MPLS offers **high security and stable performance**. Business data travels through a managed private network, making it more secure than a normal internet connection. This helps Flipkart protect customer information and ensure smooth business operations.*

Conclusion:

*Because Flipkart operates warehouses in different cities and needs secure, fast, and uninterrupted communication, **MPLS is the most suitable WAN technology** for connecting all its warehouse locations.*